

Statistical Machine Learning

1. Space of data Ω (usually is a subspace of $\mathbb{R}^m$)

In supervised learning $\Omega = X \times Y \rightarrow$ labels space
 $\uparrow$
 space of inputs

2. A model $f: X \rightarrow Y$

3. A loss function $\ell: Y \times Y \rightarrow \mathbb{R}_+$ (example $\ell(a, b) = \frac{1}{2} \|a - b\|^2$)

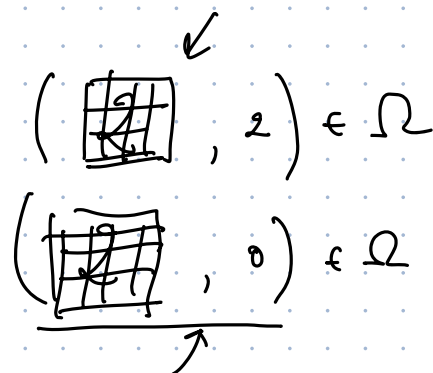
4. Probability measure μ on Ω (so that (Ω, μ) as a prob. space)

Goal is to select a model that "performs well"

Functional Risk

$$L(f) = \mathbb{E}(\ell \circ (f \times \text{id}))$$

$$= \int_{\Omega} \ell(f(x), y) d\mu(x, y)$$



"In SML you want to minimize the functional risk"

Example

$$\Omega = \mathbb{R}^2, \quad \ell(a, b) := \frac{1}{2} (a - b)^2 \quad \forall a, b \in \mathbb{R}.$$

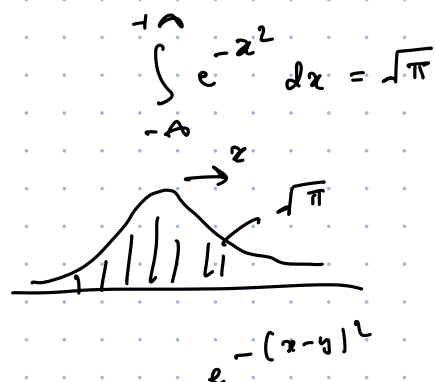
$$f: \mathbb{R} \rightarrow \mathbb{R}$$

$$\mu(A) = C \int_A e^{-(x-y)^2} e^{-x^2} dx dy$$

$$p(x, y) = C e^{-(x-y)^2} e^{-x^2}$$

$$C \int_{\mathbb{R}^2} e^{-(x-y)^2} e^{-x^2} dx dy = 1$$

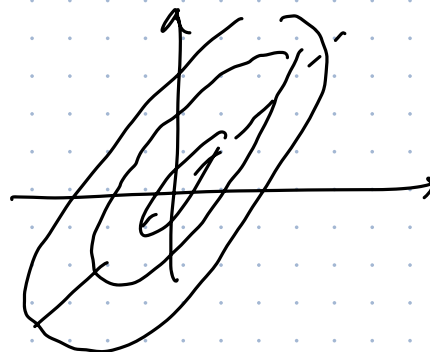
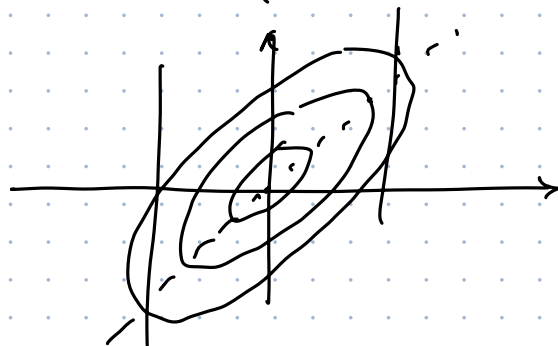
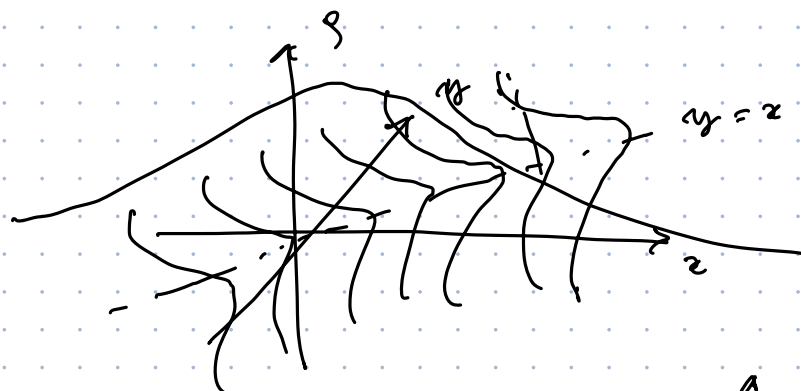
$$\underbrace{\int_{-\infty}^{\infty} dx e^{-x^2}}_{\sqrt{\pi}} \underbrace{\int_{-\infty}^{\infty} e^{-(x-y)^2} dy}_{\sqrt{\pi}}$$



$$C \cdot \pi = 1$$

$$C = \frac{1}{\pi}$$

$$p(x, y) = \frac{1}{\pi} e^{-\frac{(x-y)^2}{2}} e^{-x^2}$$



$$f(x) = x$$

This is our guess

What's the Function Risk?

$$\rightarrow L(f) = \frac{1}{2\pi} \int_{\mathbb{R}^2} (f(x) - y)^2 e^{-\frac{(x-y)^2}{2}} e^{-x^2} dx dy$$

$$e^0 \quad e^\infty$$

$$\min \{ L(f) : f \in \text{dom}(L) \}$$

$$\text{dom}(L) = \{ f \text{ measurable} : \underline{L(f) < +\infty} \}$$

$$L : \text{dom}(L) \rightarrow \mathbb{R}.$$

1. Write the first variation of L

$$\delta L(f, \sigma) = 0 \quad \forall \sigma \in V$$

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$$f \rightarrow (f_1, f_2, \dots)$$

$$f + \epsilon \sigma$$

2. We already have a candidate for minimum that is

$$f^*(x) = x$$

Try to directly prove that is a minimum

$$L(f^*) = ?$$

$$L(f) = \frac{1}{2\pi} \int_{\mathbb{R}^2} (f(x) - y)^2 e^{-(x-y)^2} e^{-x^2} dx dy$$

$$= \frac{1}{2\pi} \int_{\mathbb{R}^2} (f^2(x) + y^2 - 2yf(x)) e^{-(x-y)^2} e^{-x^2} dx dy$$

$$= \frac{1}{2\pi} \int_{\mathbb{R}^2} f^2(x) e^{-(x-y)^2} e^{-x^2} dx dy \quad (1)$$

$$+ \frac{1}{2\pi} \int_{\mathbb{R}^2} y^2 e^{-(x-y)^2} e^{-x^2} dx dy \quad (2)$$

$$- \frac{1}{\pi} \int_{\mathbb{R}^2} y f(x) e^{-(x-y)^2} e^{-x^2} dx dy \quad (3)$$

$$(1) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} dx f^2(x) e^{-x^2} \underbrace{\int_{-\infty}^{+\infty} e^{-(x-y)^2} dy}_{\sqrt{\pi}} = \frac{1}{2\sqrt{\pi}} \int_{-\infty}^{+\infty} f^2(x) e^{-x^2} dx$$

$$(3) = - \frac{1}{\pi} \int_{-\infty}^{+\infty} dx f(x) e^{-x^2} \underbrace{\int_{-\infty}^{+\infty} y e^{-(x-y)^2} dy}_{\sqrt{\pi} x} = - \frac{1}{\sqrt{\pi}} \int_{-\infty}^{+\infty} x f(x) e^{-x^2} dx$$

$$x-y = z \quad y = x-z$$

$$\int_{-\infty}^{+\infty} (x-z) e^{-x^2} dz = - \int_{-\infty}^{+\infty} z e^{-x^2} dz + x \sqrt{\pi}$$

$$(2) = \frac{1}{2\pi} \int_{-\infty}^{+\infty} dx e^{-x^2} \underbrace{\int_{-\infty}^{+\infty} dy y^2 e^{-(x-y)^2}}_{\sqrt{\pi} (x^2 + \frac{1}{2})} = \frac{1}{2\sqrt{\pi}} \int_{-\infty}^{+\infty} dx (x^2 + \frac{1}{2}) e^{-x^2}$$

$$\sqrt{\pi} x = y \quad \int_{-\infty}^{+\infty} dx (x-z)^2 e^{-x^2} = \int_{-\infty}^{+\infty} dz (x^2 + z^2 - 2xz) e^{-x^2}$$

$$-(\sqrt{\pi} x)^2 = \sqrt{\pi} x^2 + \int_{-\infty}^{+\infty} z^2 e^{-x^2} dz = \sqrt{\pi} x^2 + \frac{\sqrt{\pi}}{2}$$

$$I(a) = \int_{-\infty}^{+\infty} e^{-ax^2} dx = \frac{1}{\sqrt{a}} \int_{-\infty}^{+\infty} e^{-y^2} dy = \sqrt{\frac{\pi}{a}}$$

$$I'(a) = \int_{-\infty}^{+\infty} e^{-ax^2} (-x^2) = - \int_{-\infty}^{+\infty} x^2 e^{-ax^2} dx$$

$$I'(a) = \frac{d}{da} \left(\sqrt{\pi} a^{-1/2} \right) = -\frac{\sqrt{\pi}}{2} a^{-3/2}$$

$$I'(1) = - \int_{-\infty}^{+\infty} x^2 e^{-x^2} dx = -\frac{\sqrt{\pi}}{2}$$

$$\int_{-\infty}^{+\infty} x^2 e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$$

$$L(f) = \frac{1}{2\sqrt{\pi}} \int_{-\infty}^{+\infty} \left(f^2(x) - 2xf(x) + x^2 + \frac{1}{2} \right) e^{-x^2} dx$$

$$= \frac{1}{2\sqrt{\pi}} \int_{-\infty}^{+\infty} \left[(f(x)-x)^2 + \frac{1}{2} \right] e^{-x^2} dx$$

$$L(f^*) = \frac{1}{2\sqrt{\pi}} \int_{-\infty}^{+\infty} \frac{1}{2} e^{-x^2} dx = \frac{1}{4}$$

$$\lim_{R \rightarrow +\infty} \int_{-R}^R f(x) dx$$

What we would like to show?

$$\forall f \in \text{dom}(L) \quad L(f) \geq L(f^*) = \frac{1}{4}$$

$$L(f) = \frac{1}{2\sqrt{\pi}} \int_{-\infty}^{+\infty} \underbrace{(f(x)-x)^2}_{\geq 0} e^{-x^2} dx + \frac{1}{4} \geq 0 + \frac{1}{4} = \frac{1}{4}$$

1. In SML the assumption is that μ exists but in general it is not known!

2. Minimizing in a functional space is not so nice

1. + 2. will lead us to reformulate the FMP (in a functional space)

into the problem of minimizing the **EMPIRICAL RISK** on a given

PARAMETRIC FAMILY of functions

(2)

(1)

EMPIRICAL RISK

In general μ is not known but a sampling from μ is usually what is available in real-world problems

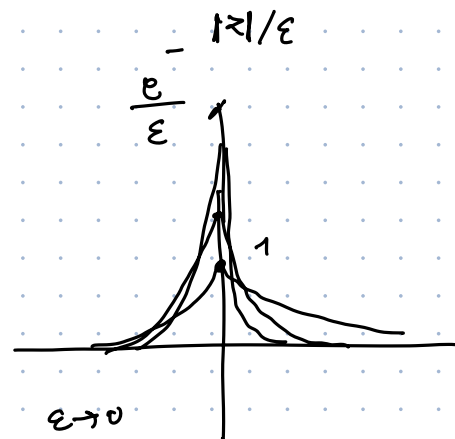
$$T = \{z_1, z_2, \dots, z_n : z_i \in \Omega\} \quad z_i = (x_i, y_i)$$

such that z_i are i.i.d. samples from μ .

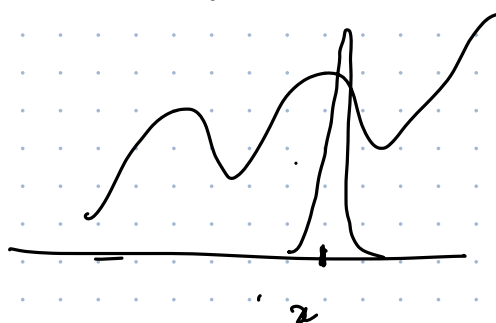
Idea is that

$$\mu \approx \frac{1}{N} \sum_{i=1}^N \delta_{(x_i, y_i)}$$

↑
"in some sense"



$$\int_{-\infty}^{+\infty} f d\delta_z = f(z)$$



$$\int_{-\infty}^{+\infty} l(f(z), y) \frac{1}{N} \sum_{i=1}^N \delta_{(x_i, y_i)}$$

$$= \frac{1}{N} \sum_{i=1}^N l(f(x_i), y_i)$$

$N \rightarrow \infty$